

7.H1 Law of Sines

Lesson Introduction

 Benchmark	MA.912.T.1.3 Apply the Law of Sines and the Law of Cosines to solve mathematical and real-world problems involving triangles.		 Learning Target	I can solve mathematical and real-world problems involving triangles using the Law of Sines.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How can I use the Law of Sines to solve problems?			
 Lesson Narrative	In this lesson, students explore the Law of Sines and how it can be used to solve both mathematical and real-world problems. Students will begin by collaborating to derive the formula for the Law of Sines. They will then learn how to use the Law of Sines as a means to solve problems.			
 Component Summary	7.H1.1 Warm-Up (~5 min.)	Notice and Wonder about two theodolites (<i>SE p. 428</i>)	7.H1.4 Guided Practice (5-10 min.)	Determining measurements using the Law of Sines (<i>SE p. 431</i>)
	7.H1.2 Collaboration (10-15 min.)	Deriving the formula for the Law of Sines (<i>SE p. 428</i>)	7.H1.5 Your Turn! (5 min.)	Determining measurements using the Law of Sines (<i>SE p. 432</i>)
	7.H1.3 Guided Instruction (10 min.)	Determining measurements using the Law of Sines (<i>SE p. 430</i>)	7.H1.6 Wrap-Up (~5 min.)	Reflecting on any difficulties or confusions in the lesson (<i>SE p. 433</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Law of Sines <u>Additional Terms:</u> N/A		Calculators	
		Additional Preparation This is an Honors lesson. Only students enrolled in Geometry Honors should be required to complete this lesson.		

 Teacher Tips	Common Misconceptions - Students may use their calculator incorrectly when finding the sine of an angle as part of the calculation. - Students may cross-multiply the original angle measure and the side length, and then find the sine of that value, instead of finding the sine of the angle first, and then cross-multiplying.	Things to Consider Students often make errors with their calculations by incorrectly using their calculator. Consider the addition of a brief tutorial or think-aloud, showing students how to find the value correctly in their calculator.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 7.H1.3 Guided Instruction provides students with explicit instruction on the Law of Sines, bridging what students discovered during the prior activity with the tasks they will complete in the next activity. Use this time to demonstrate thinking out loud and prompting students to reflect and respond to that thinking.	MA.K12.MTR.1.1: Participate in Effortful Learning 7.H1.2 Collaboration provides students with an opportunity to derive the Law of Sines. This can potentially cause frustration as it requires students to think deeply and make connections, instead of having the answer up front. Students will be encouraged to build perseverance by modifying methods, as needed, while solving a challenging task.
 Differentiate Instruction	Consider creating an anchor chart that displays the Law of Sines, along with a specific example or two of it being applied, as a reference point for students to use as they complete this lesson and the remainder of the unit. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

7.H1.1 Warm-Up: Notice and Wonder

Component Overview: Students will observe two pictures and identify what they notice and what they wonder.



7.H1.1 Warm-Up: Notice and Wonder

1. A theodolite is an instrument used for measuring angles between designated visible points on the horizontal and vertical planes. Theodolites are often used for land surveying.

Late 16 th century Theodolite	Modern Day Theodolite
	

- a. What do you notice?
Answers will vary. I notice they both sit on a base of some type.
- b. What do you wonder?
Answers will vary. I wonder if the 16th century Theodolite was as accurate as the modern day one?



This activity provides students with an image of a late 16th century theodolite, and a modern-day theodolite. They are asked to notate what they notice from the image and what they wonder from the image. This routine provides a low entry point for students to engage in general discussion.

7.H1.2 Collaboration: Deriving the Formula for Law of Sines

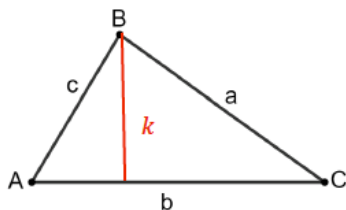
Component Overview: Students will work together to derive the formula for the Law of Sines. For this activity, consider pairing students with a partner or small group.



7.H1.2 Collaboration: Deriving the Formula for Law of Sines

Right triangle trigonometry can be used to solve problems involving right triangles, but not all triangles are right triangles. If a triangle is not a right triangle, the triangle's altitude can be used to create a right triangle so that right triangle trigonometry can be used.

1. $\triangle ABC$ is shown, where $AB = c$, $AC = b$, and $BC = a$.



- a. Use $\triangle ABC$ to do the following.

- Sketch an altitude from vertex B .
- Label the altitude k .

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle C$ is contained in the other.

- b. Complete each equation using right triangle trigonometry.

$$\sin(A) = \frac{k}{c}$$

$$\sin(C) = \frac{k}{a}$$

- c. Discuss with your partner why both equations use k . Summarize your discussion.

Answers will vary. Both equations use side k because it is shared between both triangles.



This activity provides students with an opportunity to collaborate on deriving the Law of Sines. This can potentially cause frustration as it requires students to think deeply and make connections, instead of having the answer up front. Students will be encouraged to build perseverance by modifying methods, as needed, while solving a challenging task.

7.H1.2 Collaboration: Deriving the Formula for Law of Sines (continued)

- d. Solve each equation for k .

$$k = c \sin(A), \quad k = a \sin(C)$$

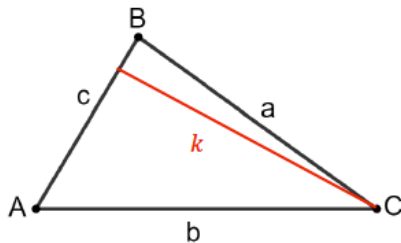
- e. Set the equations equal to each other to form a new equation.

$$c \sin(A) = a \sin(C)$$

- f. Rewrite the new equation by regrouping a with $\sin(A)$ and c with $\sin(C)$.

$$\frac{\sin(A)}{a} = \frac{\sin(C)}{c}$$

2. $\triangle ABC$ is shown where $AB = c$, $AC = b$, and $BC = a$.



- a. Use $\triangle ABC$ to do the following.

- Sketch an altitude from vertex C .
- Label the altitude k .

The altitude creates two right triangles inside $\triangle ABC$. Notice that $\angle A$ is contained in one of the right triangles, and $\angle B$ is contained in the other.



A common error in question 1d is for students to place the value of c (or a) inside the parenthesis with the sine command, creating something that looks like $k = \sin(A \cdot c)$, or $k = \sin(C \cdot a)$, which will produce the incorrect value. If you see this error, remind students of what it looks like in a non-sine environment, using a coefficient without the sine command.



Question 1f can be a confusing instructional request for students. Help them understand what this is asking, if they appear lost when trying to determine their answer.

7.H1.2 Collaboration: Deriving the Formula for Law of Sines (continued)

- b. Complete each equation using right triangle trigonometry.

$$\sin(A) = \frac{k}{b}$$

$$\sin(B) = \frac{k}{a}$$

- c. Solve each equation for k .

$$k = b \sin(A), \quad k = a \sin(B)$$

- d. Set the equations equal to each other to form a new equation.

$$b \sin(A) = a \sin(B)$$

- e. Rewrite the new equation by regrouping a with $\sin(A)$ and b with $\sin(B)$.

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b}$$



The **Law of Sines** is $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

The Law of Sines may also be written as

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



- Why is k used in both equations on question 2b?
- Is there a scenario where it would only be used in one?



Why do the two equations need to be set equal to each other?



Consider creating an anchor chart that displays the Law of Sines, along with a specific example or two of it being applied, as a reference point for students to use as they complete this lesson and the remainder of the unit.

7.H1.3 Guided Instruction: Law of Sines

Component Overview: Students will be formally introduced to the Law of Sines and its use in two problems.



7.H1.3 Guided Instruction: Law of Sines

- Some of the measurements of $\triangle DLC$ are $DL = 17$, $m\angle DLC = 40^\circ$, and $m\angle CDL = 75^\circ$. Determine the measurement of each part of $\triangle DLC$. Round to the nearest tenth.

Part	Measurement
$\overline{LC}$	$LC = 18.1$
$\overline{DC}$	$DC = 12.1$
$\angle DCL$	$m\angle DCL = 65^\circ$



This activity provides students with explicit instruction on the Law of Sines. It bridges what students discovered during the activity in the prior activity together with the tasks they will complete in the next activity. Use this time to demonstrate thinking out loud and prompting students to reflect and respond to that thinking.

- In $\triangle RFM$, $RM = 66$, $FM = 30$, and $m\angle RFM = 26^\circ$. Determine the measurement of each part of $\triangle RFM$. Round to the nearest tenth.

Part	Measurement
$\overline{FR}$	$FR = 91.6$
$\angle FRM$	$m\angle FRM = 11.5^\circ$
$\angle FMR$	$m\angle FMR = 142.5^\circ$



Do you find the use of the Law of Sines challenging? Why or why not?

7.H1.4 Guided Practice: Law of Sines

Component Overview: Students will practice two problems on the use of the Law of Sines.



7.H1.4 Guided Practice: Law of Sines

1. $\triangle VY$ has side measurements of $VY = 8.5$ and $LY = 16.5$. The measure of $\angle LVY$ is 84° . Find all missing sides and angles. Round to the nearest tenth.

$$\frac{\sin(84)}{16.5} = \frac{\sin(L)}{8.5}; \quad m\angle VLY = 30.8^\circ$$

$$180 - 30.8 - 84 = 65.2^\circ; \quad m\angle LYV = 65.2^\circ$$

$$\frac{\sin(84)}{16.5} = \frac{\sin(65.2)}{LV}; \quad LV = 15.1 \text{ units}$$

2. The three known measurements of $\triangle DNH$ are $DN = 26$, $m\angle NDH = 80^\circ$, and $m\angle DNH = 75^\circ$. Find all missing sides and angles. Round to the nearest tenth.

$$180 - 80 - 75 = 25^\circ; \quad m\angle DHN = 25^\circ$$

$$\frac{\sin(25)}{26} = \frac{\sin(80)}{NH}; \quad NH = 60.6 \text{ units}$$

$$\frac{\sin(25)}{26} = \frac{\sin(75)}{DH}; \quad DH = 59.4 \text{ units}$$



For this activity, consider breaking it up into two problems. Have students complete question 1 independently, and then bring them back together. Have each student describe a step of the process. Repeat until all steps have been shown. Engage students in a discussion to better understand any potential misconceptions before moving on to question 2.



For question 2, consider leveraging a Think-Pair-Share instructional routine. Give students a set amount of independent processing time to work through the problem on their own to see what they can do independently. Then, call "time," and have students work together as partners to calibrate their responses and come to a consensus on the correct answers, prior to reviewing as a whole class.

7.H1.5 Your Turn!

Component Overview: Students work independently to practice using the Law of Sines to solve a mathematical and a real-world problem.

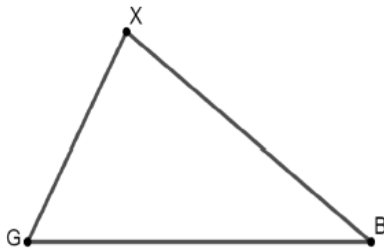


7.H1.5 Your Turn!

- In $\triangle LSK$, $LS = 7$, $m\angle KLS = 31^\circ$, and $m\angle KSL = 107^\circ$. Determine the length of $\overline{KS}$. Round to the nearest tenth.

$$\frac{\sin(31)}{KS} = \frac{\sin(42)}{7}; \quad KS = 5.4 \text{ units}$$

- Cell phone towers represented by points G and B are 25 miles apart. A new tower, X , is being built so that $m\angle G = 54^\circ$ and $m\angle B = 26^\circ$. How far from tower G will tower X be built? Round to the nearest tenth.



$$\frac{\sin(100)}{25} = \frac{\sin(26)}{GX}; \quad GX = 11.1 \text{ miles}$$



Consider using this time to pull together a small group of students who may have been struggling during the Guided Instruction or Guided Practice activities of this lesson to provide additional support with this problem in a small group setting.



How do you think a cell phone company can benefit from having the information presented in question 2?

7.H1.6 Wrap-Up: Reflection

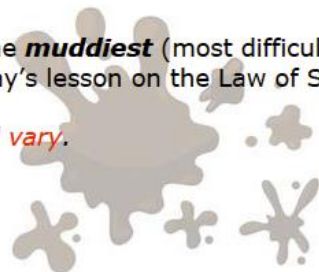
Component Overview: Students complete a self-assessment of their muddiest (most difficult or confusing) point about the Law of Sines.



7.H1.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on the Law of Sines?

Answers will vary.



Consider pairing students to discuss their reflections to see if peers can answer any of their questions.